

Turing Instabilities and Ohta–Kawasaki Phase Separation: A Dynamical Systems View of Morphogenesis

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Developmental pattern formation is often modeled either by reaction–diffusion systems that generate Turing instabilities or by phase separation models based on free energy minimization. In this work we compare these two paradigms by studying a canonical reaction–diffusion model, the Brusselator, alongside the Ohta–Kawasaki extension of the Cahn–Hilliard equation. Starting from conservation laws and flux assumptions, we derive the reaction–diffusion and phase separation equations and analyze their well-mixed limits, showing that both possess linearly stable homogeneous equilibria so that any pattern-forming instability must be spatial in origin. Linearizing the full PDEs and expanding in Fourier modes yields dispersion relations for each model, from which we identify conditions for a finite band of unstable wavenumbers and derive marginal stability curves. By matching the band edges and fastest-growing wavenumber, we construct a parameter mapping under which the two models exhibit identical onset length scales, despite their different physical interpretations. We then show that, on the center manifold, both systems reduce to the same Newell–Whitehead–Segel amplitude equation for stripe formation, so their near-onset dynamics are governed by an identical normal form. Numerical simulations confirm that the mapped models share a common instability length scale but differ in pattern type away from threshold. Together, these results demonstrate a close mathematical analogy between Turing instabilities and Ohta–Kawasaki phase separation, while highlighting how distinct microscopic mechanisms can feed into a small set of universal pattern-forming instabilities in biological and physical systems.

INTRODUCTION

Developmental biology is ultimately trying to answer a very simple-looking question: how does a single fertilized egg, with essentially uniform genetic material, reliably build a whole organism with precisely placed tissues, organs, and repeating patterns like stripes or segments? In this project we focus on one piece of that puzzle: how large-scale spatial patterns emerge, rather than on the full genetic circuitry that implements them. At the cellular level, pattern formation is driven by complex gene regulatory networks—dozens or hundreds of genes switching each other on and off in time—but for many questions about where stripes or spots appear, we don’t need to track every gene explicitly. Instead, we treat key chemical species (morphogens) as continuous fields whose production, degradation, and diffusion capture the net effect of all that underlying genetic machinery. In other words, genetics is still there, but it is “hidden” inside the reaction terms of our models, allowing us to strip away molecular detail and study the generic physical and mathematical mechanisms that can generate robust spatial patterns in developing tissues[1].

In early development cells are often genetically similar, so it has been proposed that tiny random differences in molecule numbers or local conditions are amplified by an underlying instability, causing nearby regions to diverge and organize into stripes, spots, or hexagonal patterns[2]; morphogen gradients provide exactly this kind of coordinate system. Morphogenesis is the broader process by which these local chemical cues, together with cell growth, movement, and mechanical forces, generate large-scale structure, turning a nearly uniform sheet of cells into patterned tissues, organs, and ultimately a whole organism. From this viewpoint, pattern formation is the emergence of stable spatial organization out of initially homogeneous states, driven by interactions between production, degradation, and diffusion of morphogens. A central question, both biologically and mathematically, is how relatively simple rules at the level of reaction and diffusion can robustly produce the rich variety of patterns we observe in nature. This motivates modeling morphogens and morphogenesis with partial differential equations, where we can treat patterns as steady states or dynamic attractors of an underlying reaction–diffusion system and analyze when and how spatial structure appears.

In 1951 Alan Turing proposed a new hypothesis to treat morphogenesis as a purely physico-chemical process: a small set of “morphogens” that react with each other and diffuse through tissue can, all by themselves, generate spatial patterns from an initially homogeneous state. The revolutionary step was his claim that diffusion—which had always been associated with smoothing gradients and driving systems toward uniform equilibrium—could actually destabilize a uniform chemical state when coupled to appropriate reaction kinetics. In his framework, a homogeneous equilibrium can be stable to well-mixed (spatially uniform) perturbations, yet become unstable once diffusion is included, so that tiny random fluctuations are selectively amplified into robust spatial structure (the classic “diffusion-driven instability”). This gave a concrete mechanism for how stripes, spots, or more complex patterns could emerge without pre-existing positional cues or explicit genetic prepatterns: genes set the local reaction rules (the kinetic terms), while

diffusion and instability determine the large-scale geometry of the pattern[2].

This diffusion-driven instability explanation of morphogenesis displays an incredible ability to match patterns found in biology, though the physical connection between chemical reactions and cellular motion is lacking. Recent developments in the field have sought to use diffusiophoresis, the directed motion of micro-particles up or down a concentration gradient, to bridge this gap [3], but we are interested instead in drawing mathematical analogies between the Turing Instability and other pattern-forming systems with a deeper connection to cellular motion.

Alternatively, the visible pigment patterns we study can be viewed from a phase separation perspective: instead of tracking every biochemical signal, we treat differently pigmented cell types as distinct “phases” that reorganize in space. The Differential Adhesion Hypothesis (DAH) proposes that morphogenic cells separate based on phase like mixtures of binary liquids. This leads naturally to models such as the Cahn–Hilliard equation, which forms coarsening structures that do not stabilize over time: small domains merge into larger ones, and the pattern grows without bound. This behavior stands in sharp contrast to classical Turing systems, where patterns are usually interpreted as the outcome of a competition between local activation and long-range inhibition in a reaction–diffusion system. The basic Cahn–Hilliard model lacks this long-range inhibitory mechanism, explaining its unbounded coarsening. The Ohta–Kawasaki model [4] can be viewed as an extension of Cahn–Hilliard that incorporates a nonlocal term representing long-range repulsion (for example, due to covalent bonding constraints) (in diblock copolymers). This additional term suppresses coarsening, yielding stable stripe and spot patterns that are strikingly similar to those generated by Turing instabilities (see Fig. 1). In this way, phase separation models (Cahn–Hilliard and Ohta–Kawasaki) provide an alternative, physically grounded route to pattern formation that contrasts the more mathematically driven reaction–diffusion view of Turing patterns.

The physical connection between covalent bonding in diblock copolymers and cellular interactions is not purely artificial. Experimental investigations into chromatophore interactions in zebrafish have demonstrated that similar long-range effects and nonlocal interactions are present at the cellular level [5]. At the same time, both the reaction–diffusion Turing systems and the Cahn–Hilliard–type phase separation models we study are deliberately idealized: they were originally proposed as minimal mathematical mechanisms for pattern formation, chosen because their structure lends itself to analysis (linear stability, bifurcation theory, and amplitude equations) and to the spontaneous emergence of stripes, spots, and hexagonal patterns, rather than being derived directly from detailed biological microphysics. Subsequent experimental work has helped to justify these models a posteriori, but in this project we embrace their role as simplified dynamical frameworks that distill the essential pattern-forming mechanisms. Our goal is to establish a direct mathematical analogy between the Turing instability and the Ohta–Kawasaki model, in order to lend a deeper physical backing to Turing’s original hypothesis regarding the chemical basis of morphogenesis. We will study these models near the threshold of instability and compare them locally, showing how their early-time

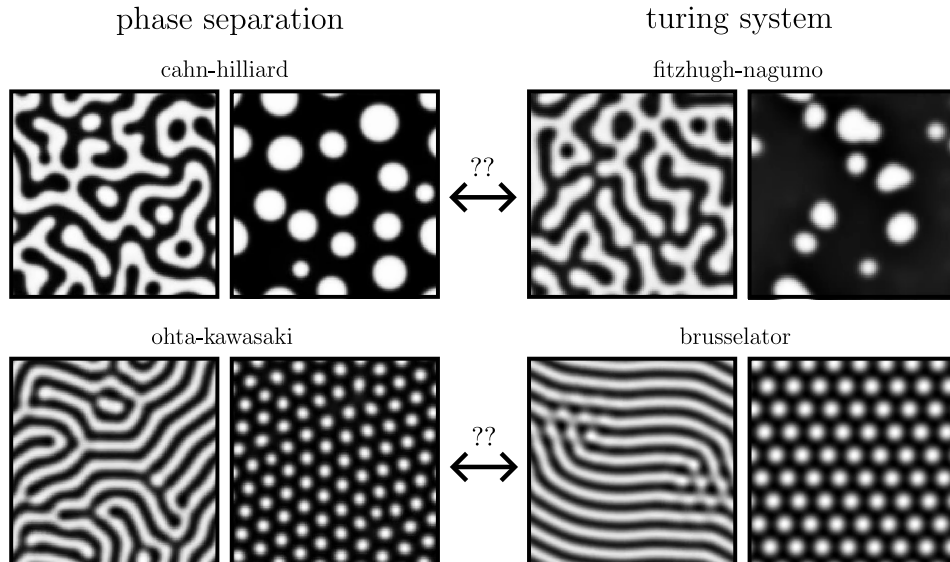


FIG. 1. A visual comparison of results for various established phase separation and Turing instability models. The cases for Cahn–Hilliard and Ohta–Kawasaki are replicated from the conditions described in [4], and the Brusselator example is replicated from [3]. FitzHugh Nagumo is created from $[D_a, D_b, \alpha, \beta, t] = [1, 100, 0.01, 0.25, 10]$ for drops and $[D_a, D_b, \alpha, \beta, t] = [1, 100, -0.005, 10, 30]$ for labyrinths. This comparison is only qualitative, but poses the question: Is there some fundamental connection between phase separations and reaction–diffusion instabilities?

dynamics can be captured by similar amplitude equations despite arising from distinct physical mechanisms.

REACTION-DIFFUSION MODELS

Diffusion models are motivated by the general conservation of mass principle, which says that the rate of change of the amount of material in a volume V is equal to the rate of flow of material across the surface S plus the material created in V .

$$\frac{\partial}{\partial t} \int_V c(\mathbf{x}, t) dV = - \int_S \mathbf{J} \cdot d\mathbf{S} + \int_V f dV$$

where $\mathbf{J}$ is the flux of the material and f is a source function. Applying the divergence theorem gives us

$$\int_V \left[\frac{\partial c}{\partial t} + \nabla \cdot \mathbf{J} - f(c, \mathbf{x}, t) \right] dV = 0$$

$$\frac{\partial c}{\partial t} + \nabla \cdot \mathbf{J} = f$$

Classical diffusion theory tells us that species diffuse from high concentrations to low concentrations, which can be mathematically represented as the flux being proportional to and directed opposite of the concentration gradient, giving us Fick's law $\mathbf{J} = -D\nabla c$ for some diffusion coefficient D . The above equation then becomes [6]

$$\frac{\partial c}{\partial t} = f + D\nabla^2 c$$

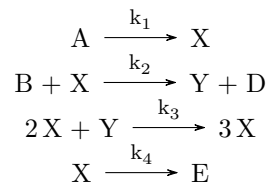
giving us the general form for reaction diffusion equations. Extending this to two chemical species, we obtain the system[1]

$$\frac{\partial X}{\partial t} = F(X, Y) + D_X \nabla^2 X$$

$$\frac{\partial Y}{\partial t} = G(X, Y) + D_Y \nabla^2 Y$$

F, G are then the reaction kinetics of the system and describe the dynamics of the chemical species without diffusion. Systems of this form are what Turing used to develop his concept of diffusion-driven instability. He considered a spatially homogeneous equilibrium (A_0, B_0) that is linearly stable in the well-mixed limit ($D_A = D_B = 0$), and then showed that when diffusion is included with unequal diffusion coefficients ($D_A \neq D_B$), this same equilibrium can become unstable to spatially varying perturbations. The resulting diffusion-driven (Turing) instability causes small random fluctuations to be amplified into spatial patterns.

Many systems of this form have been analyzed and shown to exhibit conditions leading to a Turing instability. In this project we will focus on one simple model known as the Brusselator model, which was introduced by Prigogine & Lefever [7] as a type of chemical oscillator involving source chemicals A, B , intermediate chemicals X, Y , and inert byproducts D, E . These chemicals have the following reaction scheme



It is assumed that the mixture is well stirred with constant volume and that A and B are in excess so that their concentration can be considered effectively constant. We are interested in modeling the concentrations of chemicals X and Y . The ODEs for these concentrations are then given by the law of mass action

$$\begin{aligned}\frac{dX}{dt} &= k_1 A - k_2 B X + k_3 X^2 Y - k_4 X \\ \frac{dY}{dt} &= k_2 B X - k_3 X^2 Y\end{aligned}$$

If we then nondimensionalize the system and add in the diffusion terms discussed above, we get the canonical Brusselator model

$$\begin{aligned}\partial_t X &= A - (B + 1)X + X^2 Y + \nabla^2 X \\ \partial_t Y &= B X - X^2 Y + D \nabla^2 Y\end{aligned}$$

where $D = D_Y/D_X$ is the diffusion ratio of the two species.

PHASE SEPARATION MODELS

In contrast to reaction–diffusion models that are built around the creation, movement, and destruction of multiple chemical species, phase separation models instead focus on the difference between two chemical species while conserving their total. We therefore define an order parameter $\phi(x, t)$ that describes the local difference between the fraction of cell types A and B (so that $\phi = 1$ is “all A ” and $\phi = -1$ is “all B ”), and we assume that ϕ is conserved so that the amount of material in each phase does not change, only its spatial location. Imposing the conservation law described in the previous section, we obtain

$$\frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

and phases rearrange by moving down gradients of chemical potential,

$$\mathbf{J} = -M \nabla \mu_{\text{tot}}$$

for some mobility constant $M > 0$. This gives the evolution equation

$$\partial_t \phi = M \nabla^2 \mu_{\text{tot}}$$

We now assume that the system is governed by a free energy functional $F[\phi]$ that the dynamics will try to minimize over time[8]; the free energy plays the role of a Lyapunov function: the dynamics is a gradient flow that decreases F monotonically, so in the well-mixed limit the homogeneous state is globally attracting in the sense of Lyapunov[9]. The Cahn–Hilliard construction starts from three ingredients. First, a local double–well potential $f_0(\phi)$ that favors separation of the two phases: f_0 has minima at the two pure states $\phi = \pm 1$ and a maximum at the mixed state $\phi = 0$, so that locally the system prefers to be almost all A or almost all B rather than a uniform mixture. Second, an interfacial (gradient) term

$$\frac{\kappa}{2} |\nabla \phi|^2, \quad \kappa > 0,$$

which penalizes large spatial gradients. Physically, this represents the energetic cost of interfaces: very sharp jumps between phases correspond to large $|\nabla \phi|$ and therefore high energy, so the model favors smooth transitions with a finite interface thickness rather than discontinuous jumps. Both of these contributions are local: they depend only on ϕ and its gradient at each point x , and the corresponding part of the free energy is a single spatial integral

$$\int_{\Omega} (f_0(\phi(x)) + \frac{\kappa}{2} |\nabla \phi(x)|^2) dx$$

However, the combination of a double well and a gradient penalty still favors complete phase separation of A and B into two large regions: small domains tend to merge, and the typical length scale of the pattern grows without bound

in time (coarsening). This does not produce the stable, finite-wavelength patterns we are interested in. The Ohta–Kawasaki model modifies the Cahn–Hilliard free energy by introducing a third term that penalizes large deviations from the spatial average ϕ_0 [10]

$$\frac{\alpha}{2} \int_{\Omega} \int_{\Omega} G(x-y) (\phi(x) - \phi_0) (\phi(y) - \phi_0) dx dy$$

with repulsion strength $\alpha > 0$ and a symmetric kernel $G(x-y)$. This term is nonlocal and quadratic: it sums the interaction between pairs of points x and y in the domain, weighted by $G(x-y)$. Writing it as a double integral makes clear that it represents a long-range, pairwise repulsion between regions where ϕ deviates from ϕ_0 : if a large region is strongly A -rich or B -rich, $(\phi - \phi_0)$ is nonzero over a large volume, and the double integral becomes large. In contrast, if ϕ fluctuates around its mean on shorter length scales, positive and negative contributions partially cancel and the nonlocal energy is smaller. Choosing G to depend only on $x-y$ enforces translation invariance (no preferred location in the sample).

The full Ohta–Kawasaki free energy is then

$$F[\phi] = \int_{\Omega} \left(f_0(\phi) + \frac{\kappa}{2} |\nabla \phi|^2 \right) dx + \frac{\alpha}{2} \int_{\Omega} \int_{\Omega} G(x-y) (\phi(x) - \phi_0) (\phi(y) - \phi_0) dx dy$$

In thermodynamics, the chemical potential of a species is the derivative of the free energy with respect to the number of particles: it measures the local energetic cost of adding more of that species. In a continuum setting, the analogous object is the functional derivative of F with respect to ϕ : the quantity $\delta F / \delta \phi(x)$ tells us how the free energy changes under a small local change of composition at x . We therefore define the total chemical potential field as

$$\mu_{\text{tot}}(x) = \frac{\delta F}{\delta \phi(x)}$$

For the local part of the free energy this gives a "local chemical potential" $f'_0(\phi)$ (the derivative of the double-well), and for notational convenience, we denote this by $\mu(\phi)$:

$$\mu(\phi) = f'_0(\phi)$$

Computing the functional derivative of the full $F[\phi]$ then yields

$$\mu_{\text{tot}}(x) = \mu(\phi(x)) - \kappa \nabla^2 \phi(x) + \alpha \int_{\Omega} G(x-y) (\phi(y) - \phi_0) dy.$$

To make further progress, we now choose G to be the Green's function of the Laplacian, characterized by

$$-\nabla^2 G(x-y) = \delta(x-y)$$

with appropriate boundary conditions. This choice is natural both physically and mathematically: it corresponds to a Coulomb-like long-range interaction that strongly penalizes large-scale segregation, and it allows the nonlocal term to be rewritten via a local differential operator. Applying ∇^2 to the convolution that appears in μ_{tot} and using the defining property of G , we obtain

$$\nabla^2 \int_{\Omega} G(x-y) (\phi(y) - \phi_0) dy = -(\phi(x) - \phi_0)$$

Substituting the expression for μ_{tot} into the evolution equation $\partial_t \phi = M \nabla^2 \mu_{\text{tot}}$ then gives

$$\begin{aligned} \partial_t \phi &= M \nabla^2 \left(\mu(\phi(x)) - \kappa \nabla^2 \phi(x) + \alpha \int_{\Omega} G(x-y) (\phi(y) - \phi_0) dy \right) \\ &= M \nabla^2 (\mu(\phi(x)) - \kappa \nabla^2 \phi(x)) + \alpha M \nabla^2 \int_{\Omega} G(x-y) (\phi(y) - \phi_0) dy \\ &= M \nabla^2 (\mu(\phi(x)) - \kappa \nabla^2 \phi(x)) - \alpha M (\phi(x) - \phi_0). \end{aligned}$$

Finally, absorbing the constant M into α (or, equivalently, rescaling time) we obtain the canonical form of the Ohta–Kawasaki model used in this work:

$$\partial_t \phi = M \nabla^2 (\mu(\phi) - \kappa \nabla^2 \phi) - \alpha (\phi - \phi_0)$$

The first term on the right-hand side is the Cahn–Hilliard operator, which drives phase separation and coarsening, while the second term represents a long-range restoring force that opposes separation and allows for pattern formation.

LINEAR STABILITY ANALYSIS

As described previously, systems that exhibit Turing patterns are required to have a linearly stable equilibrium when no diffusion is present: if the well-mixed equilibrium were already unstable, patterns could arise from the local reaction kinetics alone, so requiring it to be stable ensures that any instability and pattern formation is driven by diffusion. The diffusion-free Brusselator model is given by

$$\begin{aligned}\partial_t X &= A - (B + 1)X + X^2 Y, \\ \partial_t Y &= BX - X^2 Y,\end{aligned}$$

which has an equilibrium solution at $(X^*, Y^*) = (A, B/A)$. Defining new equilibrium variables $\delta X = X - X^*$ and $\delta Y = Y - Y^*$ and linearizing the system (rewriting $\delta X, \delta Y$ as X, Y again for convenience) gives

$$\begin{bmatrix} \partial_t X \\ \partial_t Y \end{bmatrix} = \begin{bmatrix} B - 1 & A^2 \\ -B & -A^2 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}.$$

The Jacobian of the linearized system therefore has trace and determinant

$$\text{tr } J = (B - 1) - A^2 = B - 1 - A^2, \quad \det J = A^2 > 0,$$

so the eigenvalues are

$$\lambda = \frac{B - 1 - A^2 \pm \sqrt{(B - 1 - A^2)^2 - 4A^2}}{2}.$$

The discriminant $(B - 1 - A^2)^2 - 4A^2$ may be positive or negative depending on A and B , so the eigenvalues can be real or complex, but in either case their real part is

$$\text{Re } \lambda = \frac{B - 1 - A^2}{2}.$$

Thus the homogeneous equilibrium is linearly stable precisely when

$$B - 1 - A^2 < 0 \iff B < 1 + A^2,$$

with a loss of stability via a Hopf bifurcation at $B = 1 + A^2$. In the classical Turing regime we therefore restrict to parameters with $B < 1 + A^2$, so that the diffusion-free Brusselator is linearly stable and any pattern-forming instability must be induced by diffusion.

In contrast, the Ohta–Kawasaki model does not have a nontrivial reaction subsystem: if we suppress all spatial variation by taking $\phi(x, t) \equiv \phi(t)$, the evolution equation reduces to

$$\dot{\phi} = -\alpha(\phi - \phi_0),$$

whose solution is $\phi(t) = \phi_0 + (\phi(0) - \phi_0)e^{-\alpha t}$, so every initial condition converges exponentially to ϕ_0 and the homogeneous equilibrium is globally stable. Thus, as in the Turing setting, the well-mixed dynamics are stabilizing rather than pattern-forming, and this shows that for both models the spatial dependence is the driving factor for instability.

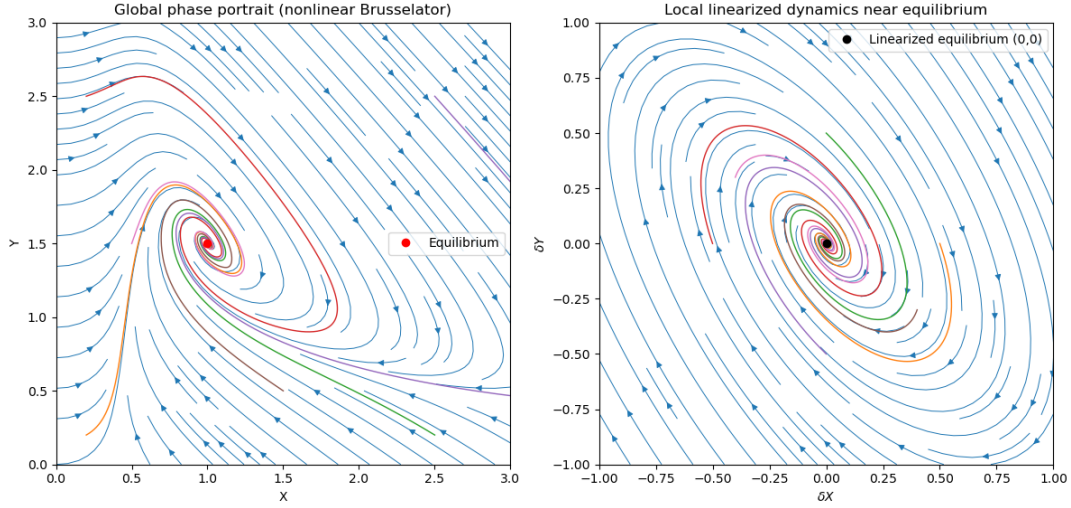


FIG. 2. Global and local dynamics of the diffusion-free Brusselator system. The left panel shows the full nonlinear phase portrait in (X, Y) for several trajectories converging to the homogeneous equilibrium (X^*, Y^*) , highlighting the globally attracting spiral behavior discussed in the reaction–diffusion section. The right panel shows the linearized dynamics in the shifted coordinates $(\delta X, \delta Y)$ near the equilibrium, illustrating that the Jacobian captures the local spiral structure used in the linear stability and bifurcation analyses.

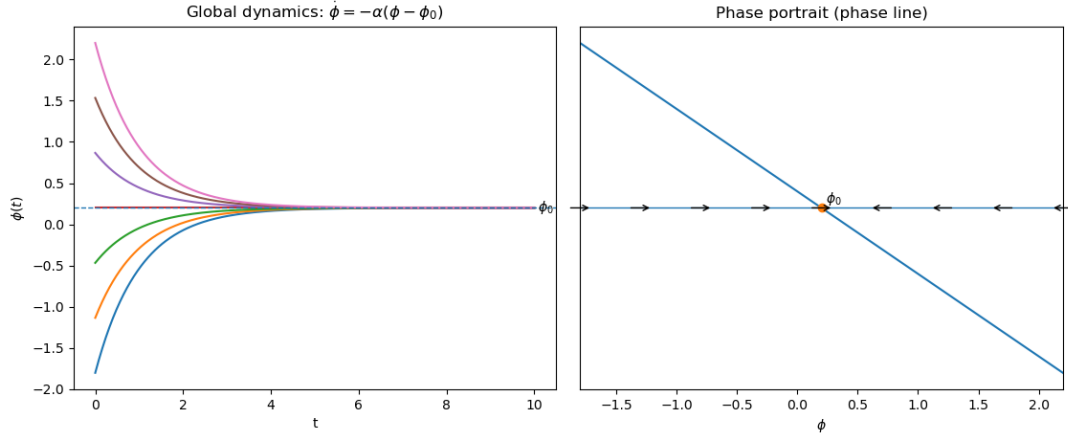


FIG. 3. Global dynamics of the well-mixed Ohta–Kawasaki model. The left panel shows solutions of $\dot{\phi} = -\alpha(\phi - \phi_0)$ for several initial conditions, all relaxing exponentially to the homogeneous state ϕ_0 . The right panel shows the corresponding one-dimensional phase line, emphasizing that ϕ_0 is a globally attracting fixed point. In contrast to the Brusselator, there is no temporal Hopf bifurcation in the ODE limit; all nontrivial structure in Ohta–Kawasaki arises purely from spatial effects.

SPATIAL INSTABILITY

Reintroducing the diffusion terms into the Brusselator system changes it from two dimensional to infinite dimensional with a continuous spectrum of eigenvalues. In finite dimensional systems the eigenvalues describe dynamics of the form $e^{\lambda t}$, so it is natural that for infinite dimensional systems with spatial dependencies we will search for dynamics of the form $e^{\lambda t + i k x}$ known as the Fourier modes with wavenumber k [11]. The linearized system is given by

$$\begin{bmatrix} \partial_t X \\ \partial_t Y \end{bmatrix} = \begin{bmatrix} B - 1 + \nabla^2 & A^2 \\ -B & -A^2 + D\nabla^2 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix},$$

and noting that $\nabla^2(e^{\lambda t + i k x}) = -k^2 e^{\lambda t + i k x}$ we obtain, along each Fourier mode,

$$\begin{bmatrix} \partial_t X \\ \partial_t Y \end{bmatrix} = \begin{bmatrix} B - 1 - k^2 & A^2 \\ -B & -A^2 - Dk^2 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}.$$

The eigenvalues $\lambda(k)$ of this 2×2 matrix satisfy

$$\lambda^2 - \tau(k)\lambda + \Delta(k) = 0,$$

where

$$\tau(k) = B - 1 - A^2 - (1 + D)k^2, \quad \Delta(k) = A^2 + A^2k^2 - Dk^2(B - 1) + Dk^4$$

are the trace and determinant, respectively. Expressing the eigenvalues as a function of the wavenumber leads to the dispersion relation

$$\lambda(k) = \frac{\tau(k) \pm \sqrt{\tau(k)^2 - 4\Delta(k)}}{2},$$

which describes how each Fourier mode of wavenumber k evolves in time. All modes with $\Re\lambda(k) < 0$ are damped and decay, whereas modes with $\Re\lambda(k) > 0$ grow exponentially; a Turing instability corresponds to a finite band of wavenumbers with $\Re\lambda(k) > 0$ while the homogeneous mode $k = 0$ remains stable.

The onset of such an instability is found by imposing the marginal condition $\lambda(k) = 0$ [12]. For a 2×2 system this is equivalent to $\Delta(k) = 0$, since $\lambda_1\lambda_2 = \Delta$ [9]. Writing $y = k^2$ reduces the quartic equation to

$$Dy^2 + (A^2 - D(B - 1))y + A^2 = 0,$$

with solutions

$$y_c = k_c^2 = \frac{D(B - 1) - A^2 \pm \sqrt{(A^2 - D(B - 1))^2 - 4DA^2}}{2D}.$$

Thus the marginal wavenumbers are

$$k_c = \left(\frac{D(B - 1) - A^2 \pm \sqrt{(A^2 - D(B - 1))^2 - 4DA^2}}{2D} \right)^{1/2},$$

and for parameter values for which the discriminant is positive these define the edges of a band of unstable modes; all k between the two positive roots have $\Re\lambda(k) > 0$ and grow to form spatial structure.

We now consider the Ohta–Kawasaki model

$$\partial_t \phi = M \nabla^2 (\mu(\phi) - \kappa \nabla^2 \phi) - \alpha(\phi - \phi_0),$$

where $\mu(\phi)$ is the local chemical potential, $\kappa > 0$ is the interfacial parameter, $M > 0$ is the mobility, and $\alpha > 0$ measures the strength of the long-range term. As in the Brusselator case, reintroducing spatial dependence changes the system from finite dimensional to infinite dimensional, since the state at time t is now the function $\phi(\cdot, t)$, and the linearized operator has a continuous spectrum of eigenvalues. In finite-dimensional systems the eigenvalues describe

dynamics of the form $e^{\lambda t}$, so for spatially extended systems with translation invariance it is natural to look for modes of the form $e^{\lambda t + i k x}$, the Fourier modes with wavenumber k . The spatial factor is written as $e^{i k x}$ with $k \in \mathbb{R}$ because these plane waves are bounded and satisfy $\nabla^2 e^{i k x} = -k^2 e^{i k x}$, so the Laplacian acts as multiplication by $-k^2$ on each mode.

This system has a homogeneous fixed point at $\phi = \phi_0$. To analyze its stability we define the equilibrium variable $\delta\phi = \phi - \phi_0$ and rewrite the equation in terms of $\delta\phi$:

$$\partial_t(\delta\phi) = M\nabla^2(\mu(\phi_0 + \delta\phi) - \kappa\nabla^2(\phi_0 + \delta\phi)) - \alpha\delta\phi.$$

Since ϕ_0 is constant in space, $\nabla^2\phi_0 = 0$, and expanding the local chemical potential to first order gives $\mu(\phi_0 + \delta\phi) \approx \mu(\phi_0) + \mu'(\phi_0)\delta\phi$. Substituting this and using that $\nabla^2\mu(\phi_0) = 0$ we obtain the linearized equation

$$\partial_t(\delta\phi) = M\nabla^2(\mu'(\phi_0)\delta\phi - \kappa\nabla^2\delta\phi) - \alpha\delta\phi.$$

For convenience we now drop the δ and write this as

$$\partial_t\phi = M\nabla^2(\mu'(\phi_0)\phi - \kappa\nabla^2\phi) - \alpha\phi.$$

As before, we look for normal modes of the form $\phi(x, t) = \hat{\phi} e^{\lambda t + i k x}$. Using $\nabla^2 e^{\lambda t + i k x} = -k^2 e^{\lambda t + i k x}$ and $\nabla^4 e^{\lambda t + i k x} = k^4 e^{\lambda t + i k x}$, substitution into the linearized equation yields the scalar dispersion relation

$$\lambda(k) = -Mk^2(\mu'(\phi_0) + \kappa k^2) - \alpha = -M\mu'(\phi_0)k^2 - M\kappa k^4 - \alpha.$$

This plays the same role as the dispersion relation $\lambda(k)$ did for the Brusselator: for each wavenumber k , the sign of $\lambda(k)$ determines whether that Fourier mode grows or decays. At $k = 0$ we have $\lambda(0) = -\alpha < 0$, so spatially uniform perturbations always decay and the homogeneous equilibrium is stable in the well-mixed sense.

To find the finite band of unstable modes we impose the marginal condition $\lambda(k) = 0$. Writing $y = k^2 \geq 0$ reduces this to a quadratic equation

$$M\kappa y^2 + M\mu'(\phi_0)y + \alpha = 0.$$

Solving for y gives

$$y_c = k_c^2 = \frac{-M\mu'(\phi_0) \pm \sqrt{M^2\mu'(\phi_0)^2 - 4M\kappa\alpha}}{2M\kappa} = \frac{-\mu'(\phi_0) \pm \sqrt{\mu'(\phi_0)^2 - \frac{4\kappa\alpha}{M}}}{2\kappa}.$$

The corresponding marginal wavenumbers are

$$k_c = \left(\frac{-\mu'(\phi_0) \pm \sqrt{\mu'(\phi_0)^2 - \frac{4\kappa\alpha}{M}}}{2\kappa} \right)^{1/2},$$

and, when the discriminant $\mu'(\phi_0)^2 - 4\kappa\alpha/M$ is positive and the resulting k_c^2 are positive, these two roots define the edges of a finite band of unstable modes. All wavenumbers k between the two positive roots have $\lambda(k) > 0$ and grow exponentially in time, while modes outside this band decay. Taking $M = 1$ recovers the commonly cited expression

$$k_c = \left(\frac{-\mu'(\phi_0) \pm \sqrt{\mu'(\phi_0)^2 - 4\kappa\alpha}}{2\kappa} \right)^{1/2},$$

which closely parallels the Brusselator result: in both models spatial dependence produces a finite-wavenumber instability of a homogeneous state, characterized by a dispersion relation and a band of growing Fourier modes that seed pattern formation.

The relationships: $\kappa = D, \alpha = A^2$, and $\partial_\phi\mu(\phi_0) = A^2 - D(B-1)$ create equality between these two band lengths. In fact, these relationships admit some physical intuition as well. The parameter κ is a penalty for interface formation, so a large κ creates diffusive interfaces, much like a large D in the Brusselator model. Here, α controls the stabilizing electrostatic force. In the linearized operator for the Brusselator model, A^2 is a bit more complex and serves two purposes. First, it couples u and v and controls the response of u , so that an increase in v corresponds to an increase in u scaled by A^2 . The second is the self-regulation of v . Note that in this case the term is $-A^2$, which is to say that if v deviates from the fixed point, it is re-stabilized by A^2 , and similarly v responds through re-stabilization. Thus,

A^2 is a stabilizing force similar to the behavior of α . The final connection is actually the most interesting, though. In this work, we care only about non-oscillatory patterns, i.e., only real eigenvalues. Consider the Brusselator dispersion relation, and note that in this case, it is unstable only for $\delta < 0$. The determinant involves $Dq^4 > 0$ and $A^2 > 0$, thus it can only be negative if the second-order term is negative, $A^2 - D(B - 1) < 0$. Of course, it must be sufficiently negative to overcome the other terms, but the sign is a necessary condition that points out that A^2 is a stabilizing term (as noted before), which must be overcome by the destabilizing term $D(B - 1)$. Looking at the Ohta-Kawasaki dispersion relation, it similarly contains three terms. The term $\kappa q^4 > 0$ is analogous to the prior diffusive term, and $\alpha > 0$ is the same A^2 term. These are once again both stabilizing. Thus, the only way to destabilize the system is for $\partial_\phi \mu(\phi_0) < 0$, to be sufficiently negative. These are the terms that control stability. This presents an interesting connection between "diffusive" stability and thermodynamic stability.

NUMERICAL COMPARISONS

In the Brusselator model, it is convenient to define $\eta = \sqrt{1/D}$. Then, the system is controlled by only two lumped parameters: 1. $A\eta$, and 2. $\gamma = B/(1 + A\eta)^2 - 1$, which arise from analyzing the onset of bifurcation [12]. We note that in the literature γ is typically denoted μ , but we make this change to avoid confusion with the chemical potential. The phase space of the Brusselator model has been well explored, and amplitude equations have been used to determine the regimes which produce stripes vs. hexagons [12]. We take an interest in mapping these parameters to the Ohta-Kawasaki model and seeing if the same pattern-type structure emerges. We can translate these parameters to the Ohta-Kawasaki model as $A\eta = \sqrt{\alpha/\kappa}$ and $\gamma = (-\partial_\phi \mu(\phi_0) - 2\sqrt{\kappa\alpha})/(\sqrt{\kappa} + \sqrt{\alpha})^2$. This only specifies two of the three parameters, so we select to hold $D = \kappa = 3.0$ constant, and allow α, ϕ_0 to vary with $A\eta, \gamma$. Typically the Ohta-Kawasaki model is defined by the double-well potential $\mu(\phi_0) = \phi_0^3 - \phi_0$, such that $\partial_\phi \mu(\phi_0) = 3\phi_0^2 - 1 \in [-1, \infty)$. We choose a parameter space for the Brusselator model which spans several pattern types, defined by: $A\eta = [1.5, 1.875]$, $\gamma \in [0.05, 0.85]$, and acknowledge that this will map to values of $\partial_\phi \mu(\phi_0) < -1$. In this specific range, the minima occurs at $A\eta = 1.875$, and $\gamma = 0.85$, where $\partial_\phi \mu(\phi_0) \approx -32.33$. We need to change the potential function, though deriving a real version which matches the behavior of the Brusselator model is certainly beyond the scope of this work (though it would be an interesting extension). We simply scale $\mu(\phi_0)$ by 35 for our analysis to keep it simple.

In Fig. 4, we demonstrate the mapped versions of the dispersion relations. It is fairly clear that the dynamics differ, specifically that the Ohta-Kawasaki model has much larger eigenvalues in every case. However, the band of unstable modes is successfully mapped as expected, and we can see that for all of the chosen sets of control parameters the mapping preserves the critical wavenumbers resulting in similar length scales between the two systems. This does not guarantee that the pattern type is properly preserved though, so we simulate these 25 cases for each model numerically in Fig. 5. We observe that indeed the pattern types do not align, and the Ohta-Kawasaki model produces almost exclusively hexagons under these conditions while the Brusselator produces a range of pattern types. The length scales do generally appear to be preserved, as is indicated by Fig. 4, though this linear mapping is clearly insufficient to fully compare the two systems. Pattern type is primarily governed by non-linear effects which are not accounted for here, and we expect that an analysis of the amplitude equations and a more complex modification to $\mu(\phi_0)$ would be necessary to fully match the pattern formation.

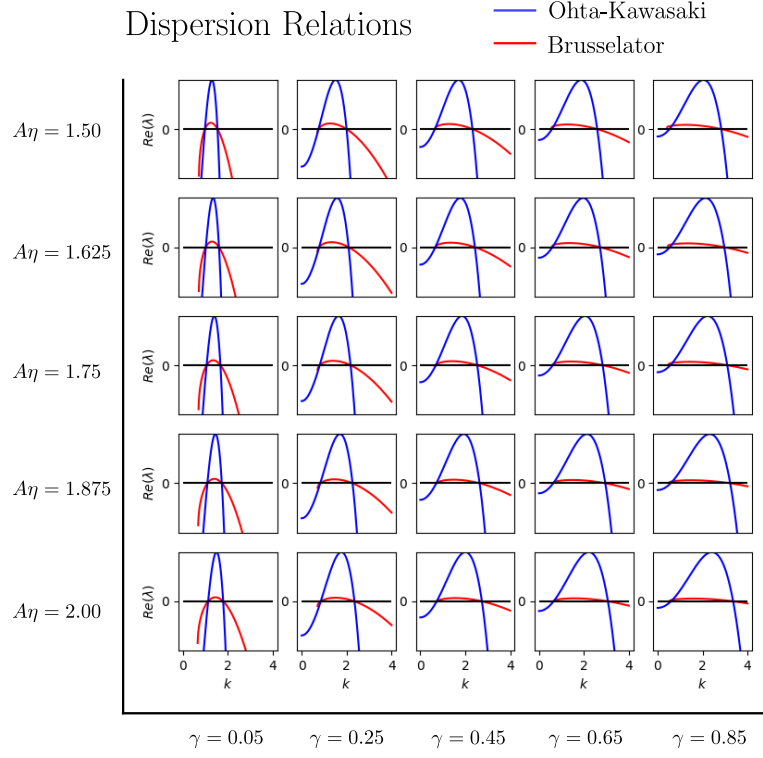


FIG. 4. Visualization of the dispersion relation for sampled points in the parameter range $A\eta \in [1.5, 2.0]$ and $\gamma \in [0.05, 0.85]$. The mapped Ohta-Kawasaki relations demonstrate the same unstable band and a similar fastest growing mode, setting the pattern length to be the same. It is clear that the dynamics differ, though the onset of instability matches.

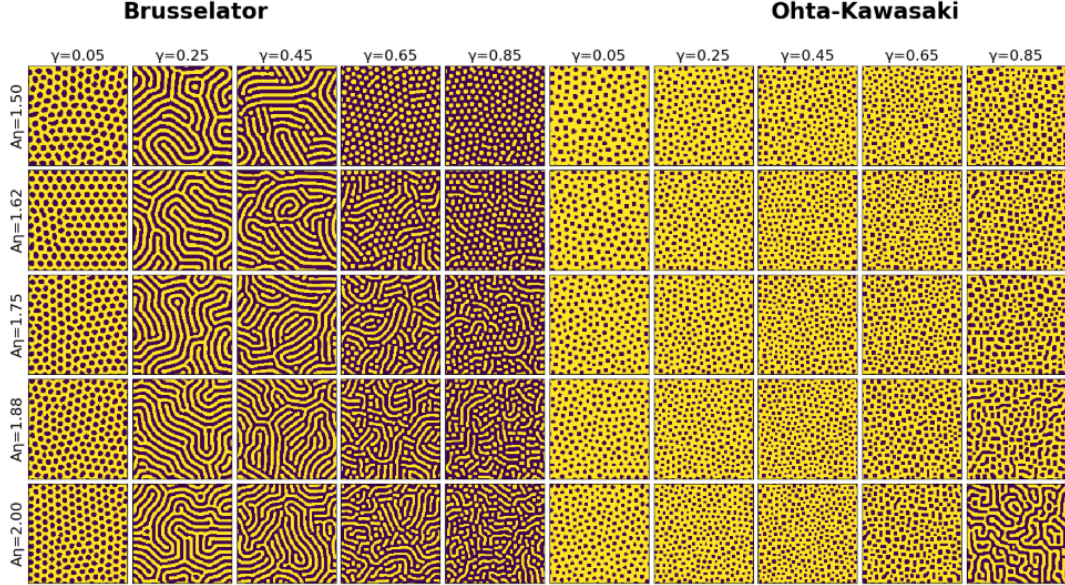


FIG. 5. Numerical results for the emergent patterns using the linear mapping. Although the length scale is properly mapped, the pattern type clearly differs, as the Ohta-Kawasaki model produces almost exclusively hexagons in this range. We expect this is a result of our over-simplified $\mu(\phi_0)$, and that a deeper investigation of non-linearity (via amplitude equations) would be required to find a potential which preserves pattern type.

BIFURCATION ANALYSIS

The Brusselator system has two bifurcations of interest: a temporal Hopf bifurcation of the well-mixed kinetics and a spatial Turing bifurcation of the reaction–diffusion system [13]. The competition between these instabilities determines whether the homogeneous steady state first loses stability to uniform oscillations or directly to a patterned state.

In the diffusion-free case ($D = 0$) we already showed that the homogeneous equilibrium $(X^*, Y^*) = (A, B/A)$ is linearly stable for a range of parameters and that the eigenvalues of the Jacobian cross the imaginary axis when the trace changes sign. This occurs at the Hopf threshold

$$B_H = 1 + A^2,$$

so that for $B < B_H$ the fixed point is a stable spiral in the well-mixed ODE system, and for $B > B_H$ it becomes an unstable spiral and the kinetics exhibit a stable limit cycle corresponding to temporal oscillations of X and Y .

When diffusion is included, the linearized reaction–diffusion system leads to the dispersion relation

$$\lambda^2 - \tau(k)\lambda + \Delta(k) = 0,$$

with $\tau(k)$ and $\Delta(k)$ as defined in the spatial instability section. A Turing instability corresponds to the situation where the homogeneous mode $k = 0$ remains stable while a finite wavenumber $k_c > 0$ satisfies $\lambda(k_c) = 0$ and becomes unstable as B is varied. Imposing $\lambda(k_c) = 0$ (equivalently $\Delta(k_c) = 0$) and minimizing the resulting marginal curve $B = B(k)$ with respect to k yields the critical values

$$(B_T, k_c) = ((1 + A\eta)^2, \sqrt{A\eta}), \quad \eta \equiv \sqrt{\frac{1}{D}},$$

so that for $B > B_T$ the homogeneous steady state is stable to spatially uniform perturbations but unstable to perturbations with wavenumber near k_c , leading to the growth of stationary spatial patterns.

For fixed A and diffusion ratio D , the first instability as B is increased is determined by comparing the Hopf threshold B_H and the Turing threshold B_T . Using the expressions above,

$$B_H = 1 + A^2, \quad B_T = (1 + A\eta)^2, \quad \eta = \sqrt{\frac{1}{D}}.$$

The Turing bifurcation occurs before the Hopf bifurcation if $B_T < B_H$, that is,

$$(1 + A\eta)^2 < 1 + A^2.$$

Expanding and simplifying gives

$$\eta^2 + \frac{2}{A}\eta - 1 < 0,$$

which holds for

$$0 < \eta < \frac{\sqrt{1 + A^2} - 1}{A}.$$

Since $\eta = \sqrt{1/D}$, this condition can be rewritten as a lower bound on the diffusion ratio D : the inhibitor Y must diffuse sufficiently faster than the activator X for the Turing instability to precede the Hopf bifurcation. In this regime the homogeneous state first loses stability to a stationary spatial pattern; if the inequality is reversed, the system instead first develops homogeneous oscillations, and spatial structure (if it appears) arises only at larger B on top of this time-periodic background.

In contrast, the Ohta–Kawasaki model does not admit a temporal Hopf bifurcation in its well-mixed limit: suppressing spatial dependence reduces the dynamics to

$$\dot{\phi} = -\alpha(\phi - \phi_0),$$

so the homogeneous state $\phi = \phi_0$ is globally stable for all parameters. When spatial dependence is included, the linear stability analysis in the previous section showed that the dispersion relation

$$\lambda(k) = -Mk^2(\mu'(\phi_0) + \kappa k^2) - \alpha$$

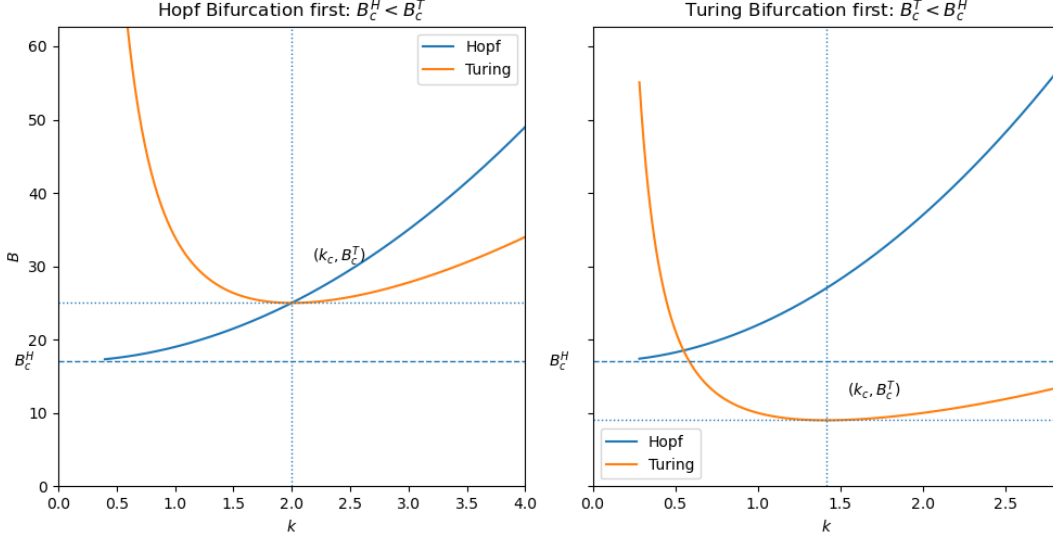


FIG. 6. Comparison of Hopf and Turing bifurcation thresholds in the Brusselator. Each panel plots the Hopf threshold $B_H = 1 + A^2$ (blue) and the marginal Turing curve $B_T(k)$ (orange) as functions of the wavenumber k . Left: a parameter regime with $B_c^H < B_c^T$, where the homogeneous state first loses stability through a temporal Hopf bifurcation. Right: a regime with $B_c^T < B_c^H$, where a finite-wavenumber Turing instability occurs first at (k_c, B_c^T) . These plots visualize the condition on the diffusion ratio D derived in the bifurcation analysis section for determining which instability is primary.

has a finite band of unstable wavenumbers whenever $\mu'(\phi_0) < 0$ and the discriminant $\mu'(\phi_0)^2 - 4\kappa\alpha/M$ is positive, with the edges of the band given by

$$k_c^2 = \frac{-\mu'(\phi_0) \pm \sqrt{\mu'(\phi_0)^2 - 4\kappa\alpha/M}}{2\kappa}.$$

In this regime the homogeneous mode $k = 0$ remains stable (since $\lambda(0) = -\alpha < 0$), but a finite range of nonzero modes grows and drives the system toward lamellar or spot-like patterns. Thus, for Ohta–Kawasaki there is only a finite-wavenumber (Turing-type) bifurcation from a globally stable homogeneous state, providing a cleaner example of purely spatial pattern formation with no competing temporal oscillatory instability.

AMPLITUDE EQUATIONS

Both of the models presented exhibit a finite band of unstable eigenmodes that propagate spatially, so the next question to consider is how these spatial instabilities evolve in time. We can analyze this by focusing on the modes right at the onset of instability, where $\lambda = 0$, and performing a center-manifold reduction to explore how the critical modes change. Expanding the solution of these systems in their critical modes [12]

$$\mathbf{u}(\mathbf{r}, t) = \sum_{j=1}^N \mathbf{u}_0(A_j(\mathbf{r}, t)e^{i\mathbf{k}_j \cdot \mathbf{r}} + \text{c.c.}) + \dots,$$

where $A_j(\mathbf{r}, t)$ is the amplitude of the j th critical mode and describes how the envelope of the mode evolves slowly in space and time. Slightly perturbing the solution and projecting onto these modes using a Fredholm-type solvability condition allows us to derive a reduced evolution equation that governs the dynamics of these amplitudes (the full derivation is outside the scope of this project). For example, the amplitude equation that describes the formation of stripes in both the Brusselator model and the Ohta–Kawasaki model can be written in the form [12, 14]

$$\partial_t A = \mu A - g|A|^2 A + \varepsilon_0^2 \left(\partial_x + \frac{1}{2ik_c} \partial_y^2 \right)^2 A,$$

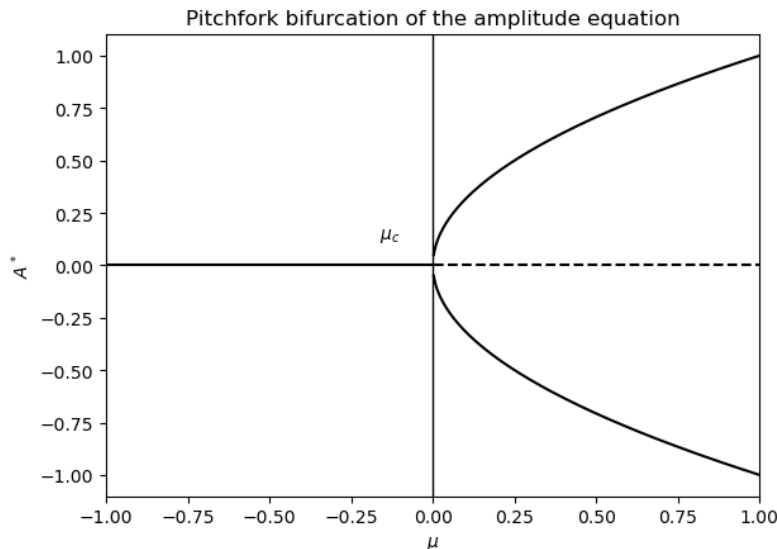


FIG. 7. Supercritical pitchfork bifurcation of the stripe amplitude equation. The stationary solutions A^* of the reduced amplitude dynamics are plotted as a function of the control parameter μ , showing the trivial homogeneous state $A^* = 0$ (dashed) losing stability at $\mu = 0$ and a pair of stable finite-amplitude branches $A^* = \pm\sqrt{\mu/g}$ emerging for $\mu > 0$. This normal-form structure underlies both the Brusselator and Ohta–Kawasaki models near the onset of pattern formation and explains why they exhibit similar stripe amplitudes close to threshold despite their different microscopic origins.

which is the normal form for a supercritical bifurcation plus a term that accounts for slow spatial variations, known as the Newell–Whitehead–Segel equation. Ignoring the spatial derivative term, the amplitude equation reduces to $\dot{A} = \mu A - g|A|^2 A$, whose radial dynamics $\dot{R} = \mu R - gR^3$ describe a supercritical bifurcation from the homogeneous state ($R = 0$) to a branch of stable finite-amplitude stripes ($R = \sqrt{\mu/g}$ for $\mu > 0$). The derivative operator then captures how these stripes can slowly modulate, bend, or adjust their wavelength in space.

The fact that both models reduce to the same amplitude equation means that, near onset, their dynamics is topologically equivalent to the same normal form—an infinite-dimensional analog of the Hartman–Grobman theorem from finite-dimensional systems. Both models collapsing to the same single-amplitude dynamics on the center manifold explains why such similar patterns emerge at the onset of instability despite being derived from entirely different contexts and exhibiting drastically different global dynamics.

CONCLUSION

The comparison between reaction–diffusion and phase separation models highlights a surprising mixture of difference and universality in pattern formation. The Brusselator treats morphogens as reacting and diffusing species that destabilize a homogeneous equilibrium through a diffusion-driven (Turing) instability, while the Ohta–Kawasaki model treats pigment or cell types as conserved phases evolving under a free energy with local and nonlocal contributions. Yet, once linearized and expressed in Fourier space, both frameworks exhibit a finite band of unstable modes and an almost identical dispersion relation, and with an appropriate parameter mapping the band edges and fastest-growing wavenumber coincide. On the center manifold, both models further reduce to the same Newell–Whitehead–Segel amplitude equation for stripe formation, so near onset the envelope dynamics of the critical mode are governed by an identical normal form. This shared structure explains why such different microscopic stories can generate spatially similar stripes and spots on comparable length scales.

At the same time, the differences between the two descriptions carry important biological implications. Reaction–diffusion systems like the Brusselator generally support both temporal and spatial instabilities; the presence of a Hopf bifurcation in the well-mixed kinetics makes homogeneous oscillations and time-dependent patterns natural, matching settings where gene expression and morphogen concentrations pulse in time as well as space [6]. By contrast, the Ohta–Kawasaki model has a trivially relaxing well-mixed limit and only a finite-wavenumber instability once spatial dependence is included, aligning more naturally with conserved, energy-driven rearrangements such as cell sorting or microphase separation. Empirically, Turing-type reaction–diffusion ideas have seen their clearest suc-

cesses in synthetic or engineered systems, where gene circuits and chemical networks have been designed to exhibit diffusion-driven instabilities, while Ohta–Kawasaki-like physics has direct biological support in systems such as zebrafish chromatophores, where long-range, nonlocal interactions between pigment cells resemble the nonlocal term in the free energy [2, 5]. This sits within a broader landscape in which many mechanisms — differential adhesion, chemotaxis, mechanochemical buckling, and growth-induced instabilities — can all generate spatial patterns, suggesting that similar macroscopic motifs (stripes, spots, labyrinths) need not arise from a unique microscopic cause.

Taken together, these observations caution against inferring mechanism directly from pattern while also offering a constructive way forward. A finite band of unstable modes and a characteristic wavelength can be produced by both reaction–diffusion instabilities and Ohta–Kawasaki-type phase separation, so additional signatures are essential to distinguish between competing hypotheses in a given biological system. At the same time, the close match between dispersion relations and amplitude equations suggests that different microscopic mechanisms may often be interchangeable at the level of reduced models: multi-mode amplitude equations, hybrid reaction–diffusion/phase-separation descriptions, and data-driven fitting of effective dispersion relations to experimental movies are natural next steps. Rather than elevating a single “correct” mechanism of morphogenesis, the mathematical analogy between Turing and Ohta–Kawasaki systems points toward a unifying view in which diverse biochemical, mechanical, and energetic processes feed into a small number of universal pattern-forming instabilities, with biology selecting among them — or combining them — in context-dependent ways.

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